Movie Ticket Booking System (Python Web Application)

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ABSTRACT

The Movie Ticket Booking System is a Python-based mini project designed to automate and simplify the process of booking movie tickets. The system allows users to view available movies, show timings, ticket prices, and seat availability in an organized manner. It provides an easy and efficient way for customers to select their preferred movie and book tickets without manual intervention.

This application is developed using Python programming concepts such as functions, conditional statements, loops, object-oriented programming, and file handling. The system maintains records of movies including movie name, show time, screen number, ticket price, and available seats. Users can select a movie, choose the number of tickets, and confirm the booking. Once the booking is completed, the system automatically updates seat availability and generates a booking confirmation.

The project helps in reducing booking errors, saving time, and maintaining accurate records of transactions. Booking data can be stored using file handling or integrated with a database for better data management and scalability.

This mini project demonstrates the practical application of Python in developing real-world automation systems and serves as a foundation for building advanced booking platforms with graphical user interfaces and online payment integration.

PROBLEM DESCRIPTION

Traditional movie ticket booking methods often involve manual ticket counters, long queues, and inefficient seat management. This can lead to booking errors, inconvenience for customers, and difficulty in managing theatre seat availability.

The objective of this project is to develop a web-based Movie Ticket Booking System that allows users to easily browse movies and book tickets online. The system should provide a convenient interface where users can view movie posters, check show timings, select theatres, and choose seats from a visual theatre layout.

The system focuses mainly on Malayalam and Indian movies and aims to simulate the working of modern online booking platforms such as BookMyShow. The project demonstrates how Python programming can be used to automate booking operations, manage movie data, and provide a better user experience.

LIBRARIES & TOOLS USED

Library / Tool	Purpose
Python	Main programming language used for backend development
Flask	Web framework used to build the backend server and handle routes
HTML	Used to create the structure of web pages
CSS	Used to design and style the user interface
JavaScript	Used to implement dynamic seat selection and interactive features
SQLite	Used to store movie data, theatre details, and booking records
VS Code	Development environment used to write and run the project

SYSTEM DESIGN / WORKFLOW

The Movie Ticket Booking System follows a simple workflow to allow users to browse and book movie tickets efficiently.

Step 1 – Home Page:
The home page displays a list of movies with their posters. Users can browse Malayalam and Indian movies.

Step 2 – Search Movies:
A search bar is provided where users can search for movies by name.

Step 3 – Movie Details:
When a user selects a movie, the system displays details such as:

- Movie poster
- Description
- Language
- Duration
- Available theatres.

Step 4 – Theatre Selection:
Users can select from multiple theatres and view different show timings.

Step 5 – Seat Selection:
A theatre-style seat layout is displayed where users can select available seats.

Step 6 – Booking Confirmation:
After selecting seats, the system confirms the booking and updates seat availability.

IMPLEMENTATION DETAILS

The system is implemented using Python Flask for the backend and HTML, CSS, and JavaScript for the frontend.

Backend (Python Flask)
The backend handles:

- Routing between pages
- Movie data management
- Booking logic
- Seat availability updates

Example routes include:

- / → Displays home page with movies
- /movie/<id> → Shows movie details
- /theatre/<movie> → Displays theatre options
- /seats → Seat selection page
- /confirm → Booking confirmation page

Frontend
The frontend consists of several web pages:

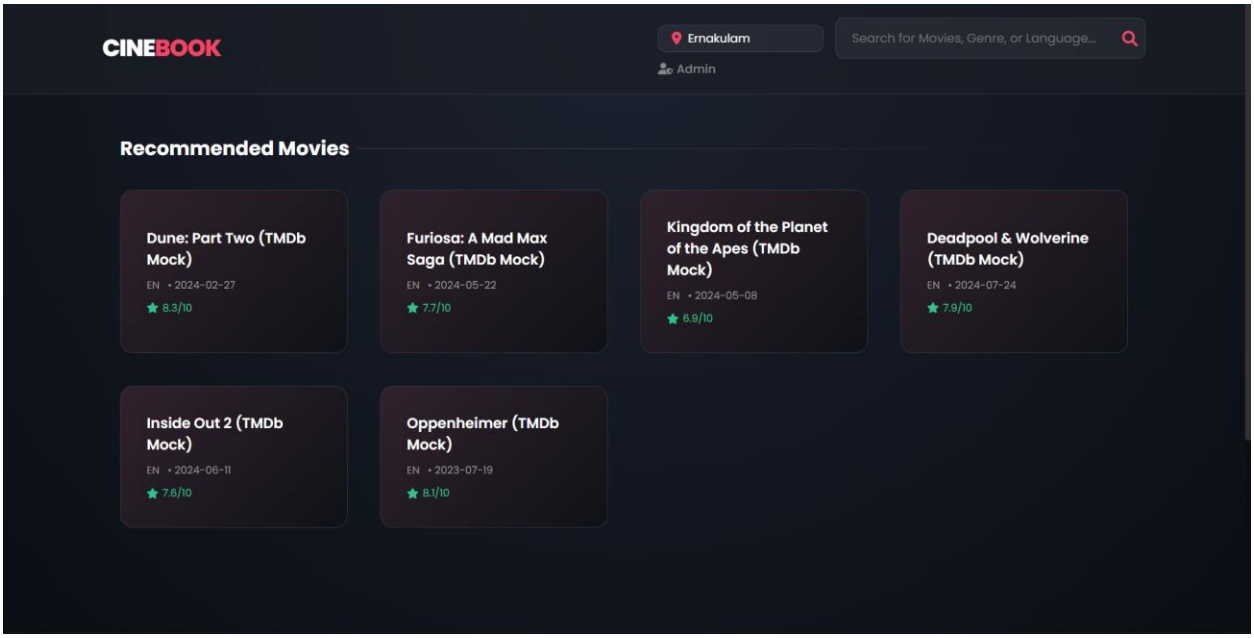
1. Home Page
   Displays movie posters with "Book Now" buttons.
2. Movie Details Page
   Shows movie information and available theatres.
3. Theatre Page
   Displays show timings for different theatres.
4. Seat Selection Page
   Shows a theatre-style seat layout where users can select seats.
5. Confirmation Page
   Displays booking confirmation with selected seats.

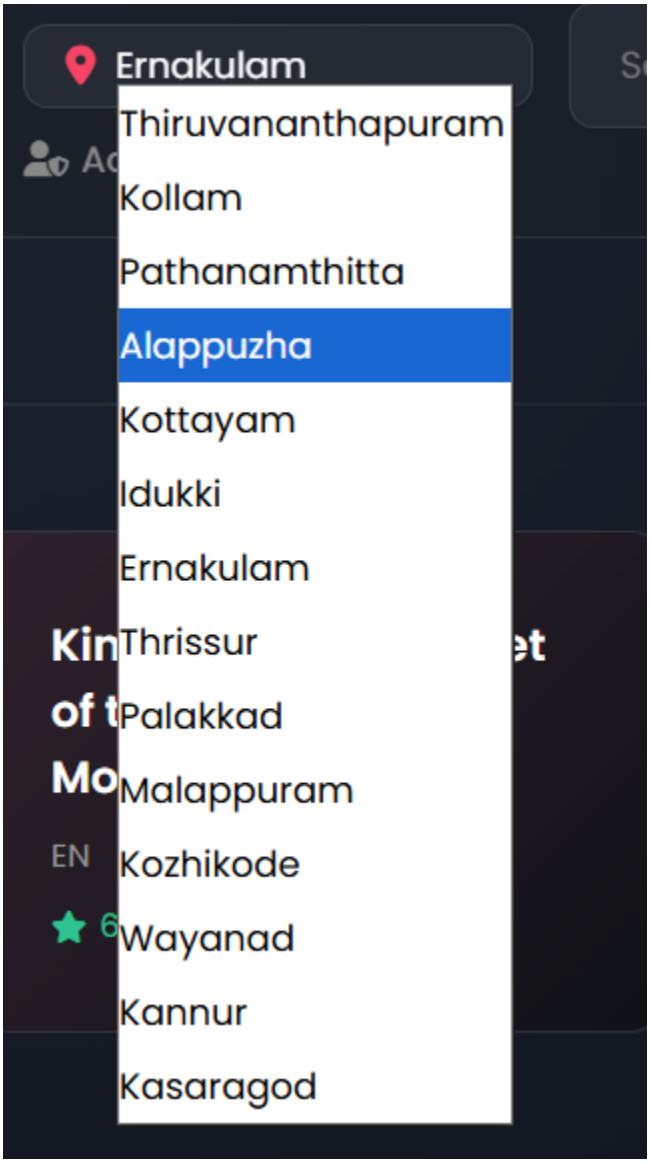
The seat layout is designed using HTML grid layout and JavaScript to allow users to select seats interactively.

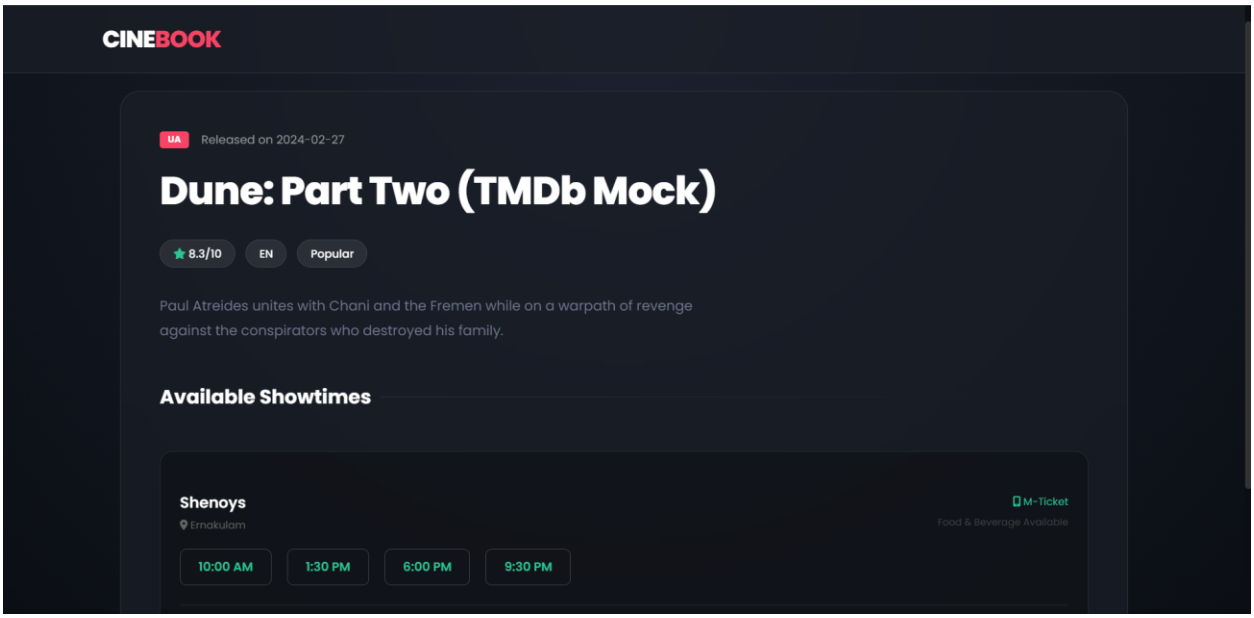
RESULTS & OUTPUT (WITH SCREENSHOTS)

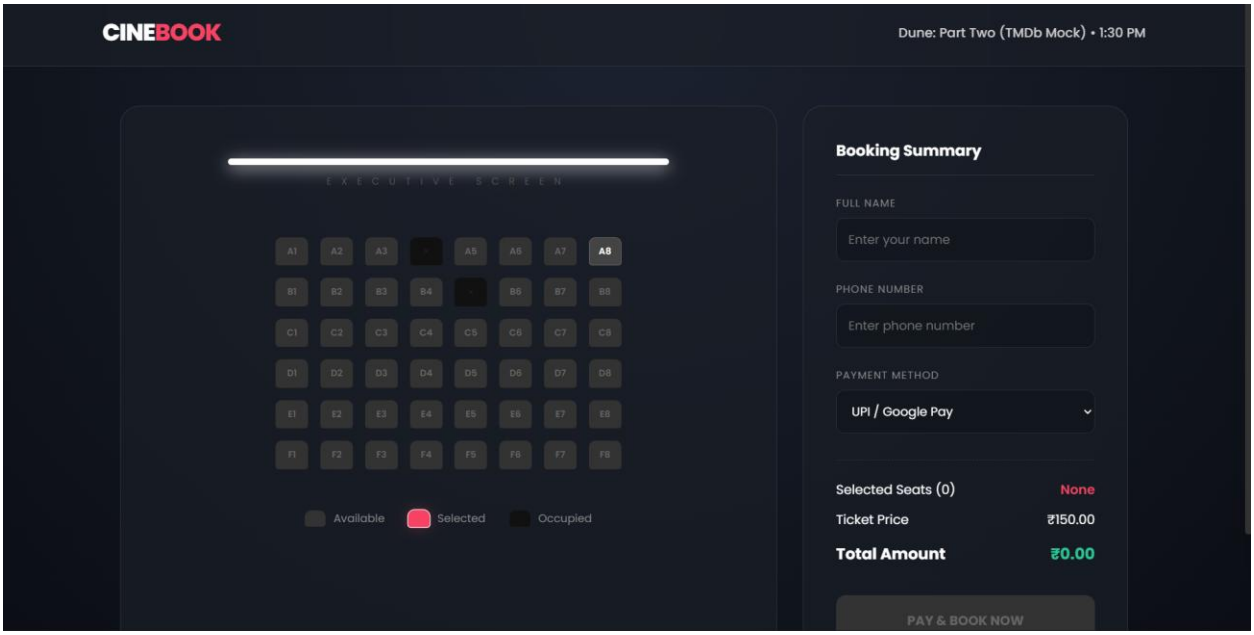
The system was successfully implemented and tested. The application allows users to browse movies and book tickets easily through the website interface.

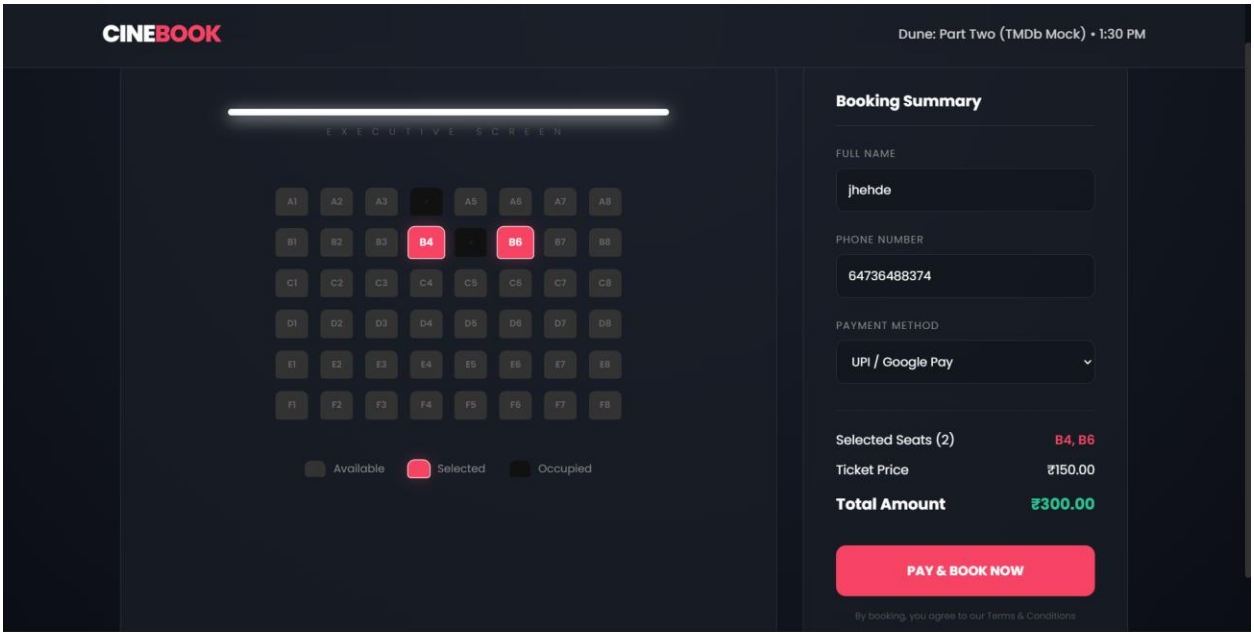
Output Screenshots to Include

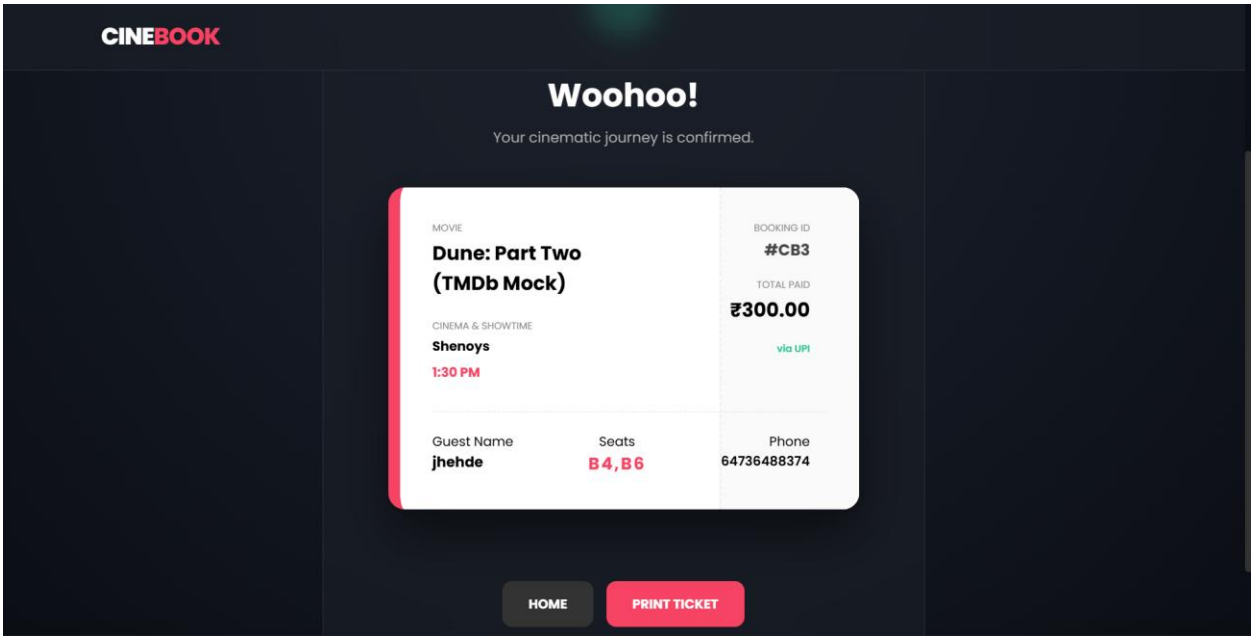


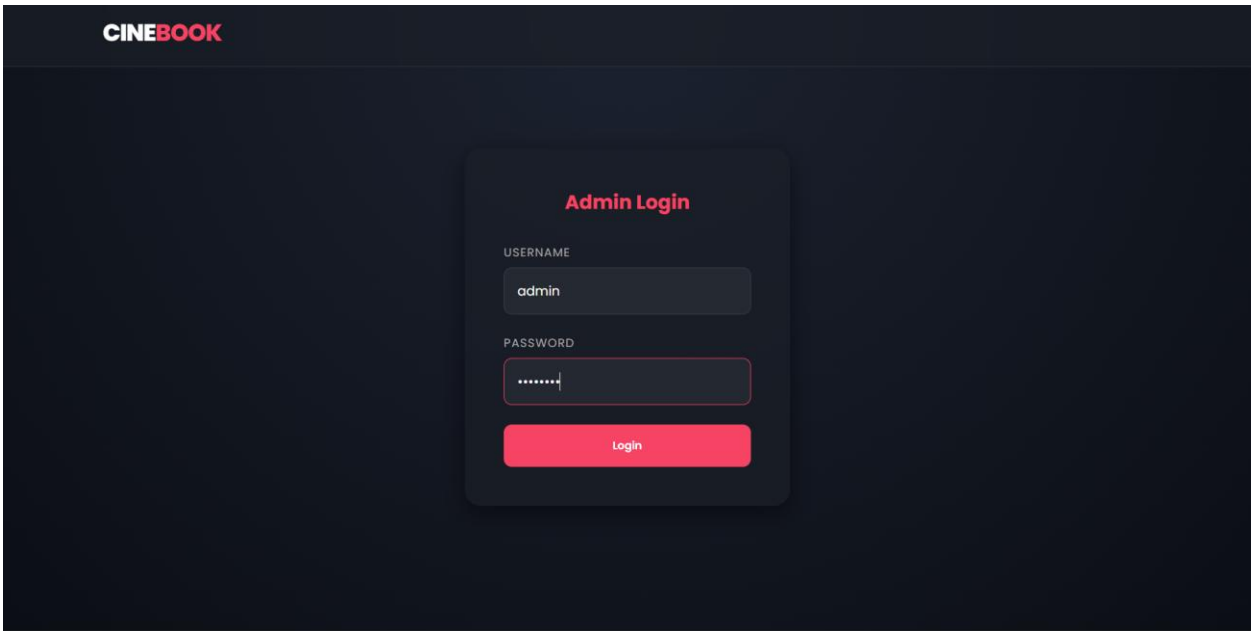

Ernakulam
Thiruvananthapuram
Kollam
Pathanamthitta
Alappuzha
Kottayam
Idukki
Ernakulam
Thrissur
Palakkad
Malappuram
Kozhikode
Wayanad
Kannur
Kasaragod
EN

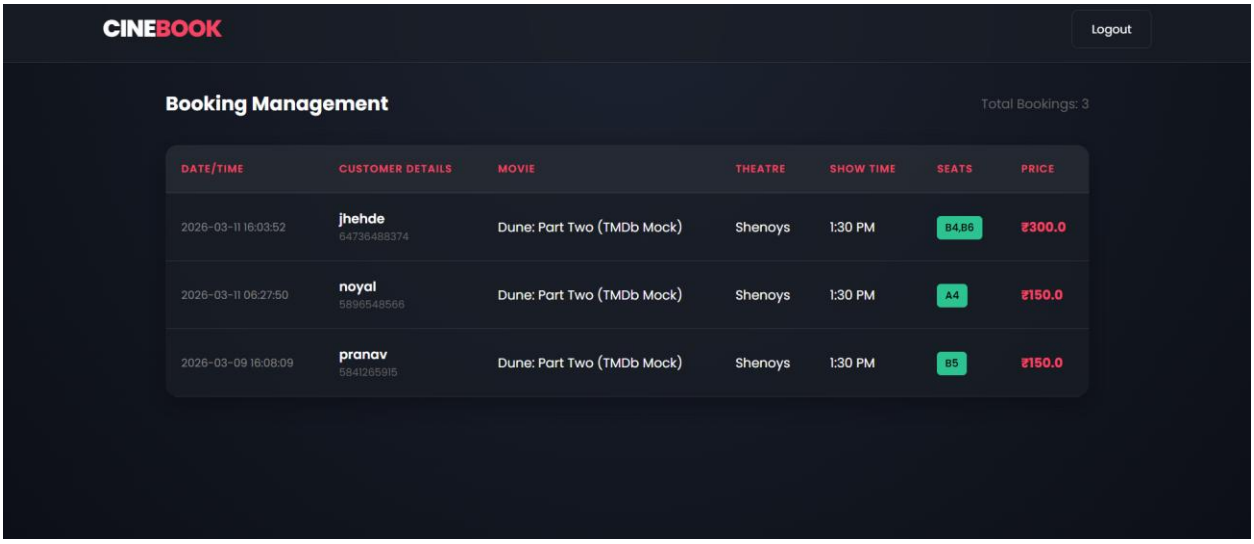

CINEBOOK
UA
Released on 2024-02-27
Dune: Part Two (TMDb Mock)
8.3/10
EN
Popular
Paul Atreides unites with Chani and the Fremen while on a warpath of revenge against the conspirators who destroyed his family.
Available Showtimes
Shenoys
Ernakulam
M-Ticket
Food & Beverage Available
10:00 AM
1:30 PM
6:00 PM
9:30 PM


CINEBOOK
Dune: Part Two (TMDb Mock) • 1:30 PM
EXECUTIVE SCREEN
A1 A2 A3 A5 A6 A7 A8
B1 B2 B3 B4 B6 B7 B8
C1 C2 C3 C4 C5 C6 C7 C8
D1 D2 D3 D4 D5 D6 D7 D8
E1 E2 E3 E4 E5 E6 E7 E8
F1 F2 F3 F4 F5 F6 F7 F8
Available
Selected
Occupied
Booking Summary
FULL NAME
Enter your name
PHONE NUMBER
Enter phone number
PAYMENT METHOD
UPI / Google Pay
Selected Seats (0)
None
Ticket Price
₹150.00
Total Amount
₹0.00
PAY & BOOK NOW


CINEBOOK
Dune: Part Two (TMDb Mock) • 1:30 PM
EXECUTIVE SCREEN
A1 A2 A3 A5 A6 A7 A8
B1 B2 B3 B4 B6 B7 B8
C1 C2 C3 C4 C5 C6 C7 C8
D1 D2 D3 D4 D5 D6 D7 D8
E1 E2 E3 E4 E5 E6 E7 E8
F1 F2 F3 F4 F5 F6 F7 F8
Available
Selected
Occupied
Booking Summary
FULL NAME
jhehde
PHONE NUMBER
64736488374
PAYMENT METHOD
UPI / Google Pay
Selected Seats (2)
B4, B6
Ticket Price
₹150.00
Total Amount
₹300.00
PAY & BOOK NOW
By booking, you agree to our Terms & Conditions


CINEBOOK
Woohoo!
Your cinematic journey is confirmed.
MOVIE
Dune: Part Two (TMDb Mock)
CINEMA & SHOWTIME
Shenoys
1:30 PM
BOOKING ID
#CB3
TOTAL PAID
₹300.00
via UPI
Guest Name
jhehde
Seats
B4,B6
Phone
64736488374
HOME
PRINT TICKET


CINEBOOK
Admin Login
USERNAME
admin
PASSWORD
Login


CINEBOOK
Logout
Booking Management
Total Bookings: 3
DATE/TIME
CUSTOMER DETAILS
MOVIE
THEATRE
SHOW TIME
SEATS
PRICE
2026-03-11 16:03:52
jhehde
64736488374
Dune: Part Two (TMDb Mock)
Shenoys
1:30 PM
B4,B6
₹300.0
2026-03-11 06:27:50
noyal
5896548566
Dune: Part Two (TMDb Mock)
Shenoys
1:30 PM
A4
₹150.0
2026-03-09 16:08:09
pranav
5841265915
Dune: Part Two (TMDb Mock)
Shenoys
1:30 PM
B5
₹150.0

Conclusion & Scope for Mini Project Extension

The Movie Ticket Booking System successfully demonstrates the use of Python for developing a web-based application. The system provides a user-friendly interface where users can browse movies, check show timings, select theatres, and book seats efficiently.

The project helped in understanding web development concepts such as backend routing, frontend design, and data management. It also demonstrates the practical use of Python libraries such as Flask for building real-world applications.

Future Improvements:
The system can be further improved by adding:

- User login and registration
- Online payment integration
- Admin panel to manage movies
- Real-time seat availability
- Email ticket confirmation
- Mobile responsive design

These enhancements can transform the mini project into a complete movie booking platform.